Project: Worked examples
Title: Tolerance stack-up (1D)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the worst-case, root-sum-square and modified RSS limits of a linear tolerance loop, with the predicted spread and capability. It adds a contribution Pareto and a Monte Carlo prediction against a stated requirement.

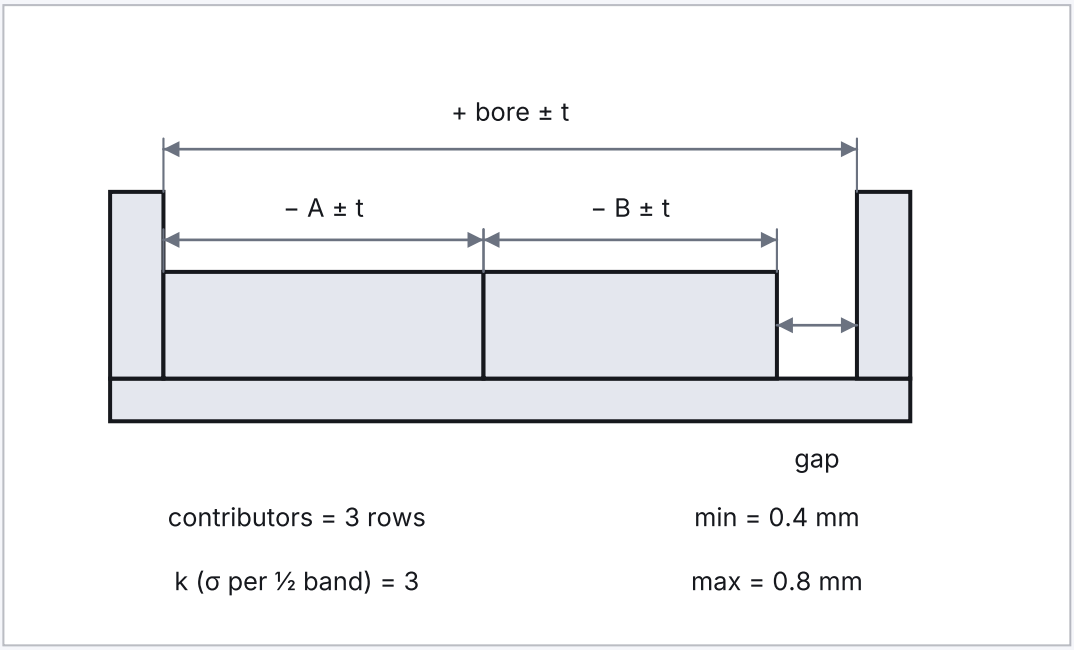


Figure 1 — A linear stack (representative; the contributors as listed): the bore adds, the parts subtract, the gap closes the loop. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Worst case, every contributor at its limit	$T_{wc} = \Sigma\lvert\text{half-bands}\rvert$
Root sum of squares, the statistical stack	$T_{rss} = \sqrt{\Sigma}\ \text{half}^2$
Modified RSS, a factor on the RSS (Bender-style)	$T_{mrss} = \text{factor} \times \text{RSS}$
Predicted standard deviation of the stack, from each contributor's distribution	$\sigma = \sqrt{\Sigma}\ \sigma_i^2$
Process capability, taken from the nearer limit	$C_{pk} = \min(\text{USL} - \mu, \mu - \text{LSL})/(3\sigma)$
Monte Carlo check, every contributor sampled from a fixed seed and summed with its sign	stack = $\Sigma\ s_i\ x_i$

Basis: Technical product documentation and specification[1]; Dimensioning and tolerancing[2]; General tolerances (where a contributor's band is a general tolerance)[3].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
contributors	Contributors in the loop	3 row(s)		drawing: DRG-HSG-001 rev B	any
notation	Drawing notation	BS 8888		assumption: Metric drawing.	any
k	Standard deviations per half-band (normal contributors)	3		assumption: Each part at ±3σ within its band.	1 … 6
f_{rss}	Modified-RSS factor (Bender-style)	1.5		assumption: Bender factor 1.5.	1 … 2
cpk_{req}	Required C_{pk}	1.33		assumption: Usual capability target.	1 … 2
minimum	Minimum acceptable result (optional; blank = no minimum)	0.4	mm	spec: SPEC-BRG-7 §4.2	-10000 … 10000 mm
maximum	Maximum acceptable result (optional; blank = no maximum)	0.8	mm	spec: SPEC-BRG-7 §4.2	-10000 … 10000 mm
n_{mc}	Monte Carlo samples	10000		assumption: Default sample count.	1000 … 100000
seed	Monte Carlo seed	1		assumption: Fixed seed for reproducibility.	0 … 1000000000

TABLE ROW	VALUES
contributors[1]	name = Housing bore depth; direction = adds; nominal = 40 mm; upper = 0.1 mm; lower = 0 mm
contributors[2]	name = Bearing width; direction = subtracts; nominal = 20 mm; upper = 0 mm; lower = -0.12 mm
contributors[3]	name = Spacer; direction = subtracts; nominal = 19.5 mm; upper = 0.02 mm; lower = -0.02 mm

3. Assumptions

A1. Every contributor acts along the one axis of the stack. (A linear stack: no angular or datum-shift effects beyond what the user enters as a contributor.)
A2. Contributors vary independently about their mid-limit and any stated mean shift, each with its stated distribution. (The worst case is a guarantee; every statistical figure is a prediction about a population.)
A3. A position or profile contributor contributes ± half its tolerance zone, plus half the stated MMC bonus. (Diametral zones are bilateral about the true position.)
A4. notation (Drawing notation) is taken as BS 8888. (Metric drawing.)
A5. k (Standard deviations per half-band (normal contributors)) is taken as 3. (Each part at ±3σ within its band.)
A6. f_{rss} (Modified-RSS factor (Bender-style)) is taken as 1.5. (Bender factor 1.5.)
A7. cpk_{req} (Required C_{pk}) is taken as 1.33. (Usual capability target.)
A8. n_{mc} (Monte Carlo samples) is taken as 10000. (Default sample count.)

A9. seed (Monte Carlo seed) is taken as 1. (Fixed seed for reproducibility.)

4. Working

[7] c1 = + 40 +0.1/0
= mid 40.05 mm, ± 0.05 mm, σ 0.0166667 mm (normal)
= 40.05 mm
Housing bore depth

[8] c2 = - 20 +0/-0.12
= mid 19.94 mm, ± 0.06 mm, σ 0.02 mm (normal)
= -19.94 mm
Bearing width

[9] c3 = - 19.5 ±0.02
= mid 19.5 mm, ± 0.02 mm, σ 0.00666667 mm (normal)
= -19.5 mm
Spacer

[10] x_{nom} = Σ (sign × nominal)
= 40 - 20 - 19.5
= 0.5 mm
Nominal result

[11] x_{mean} = Σ (sign × mid-limit)
= 40.05 - 19.94 - 19.5
= 0.61 mm
Mean result

[12] t_{wc} = Σ half-bands
= 0.05 + 0.06 + 0.02
= 0.13 mm
Worst-case tolerance (±)

[13] $x_{wc,min}$ = x_{mean} - t_{wc}
= 0.61 mm − 0.13 mm
= 0.48 mm
Worst-case minimum

[14] $x_{wc,max}$ = x_{mean} + t_{wc}
= 0.61 mm + 0.13 mm
= 0.74 mm
Worst-case maximum

[15] t_{rss} = √Σ half²
= √(0.05² + 0.06² + 0.02²)
= 0.080623 mm
Root-sum-square tolerance (±)

[16] $x_{rss,min}$ = x_{mean} - t_{rss}
= 0.61 mm − 0.080623 mm
= 0.52938 mm
RSS minimum

[17] $x_{rss,max}$ = x_{mean} + t_{rss}
= 0.61 mm + 0.080623 mm
= 0.69062 mm
RSS maximum

[18] t_{mrss} = f_{rss} * t_{rss}
= 1.5 × 0.080623 mm
= 0.12093 mm
Modified RSS tolerance (±)

[19] σ = √Σ σ_i^2
= √(0.0166667² + 0.02² + 0.00666667²)
= 0.026874 mm
Predicted standard deviation of the result

[20] m_{min} = $x_{wc,min}$ - minimum
= 0.48 - 0.4
= 0.08 mm
Worst-case margin to the minimum
[21] m_{max} = maximum - $x_{wc,max}$
= 0.8 - 0.74
= 0.06 mm
Worst-case margin to the maximum
[22] cpk = min(x_{mean} - LSL, USL - x_{mean}) / (3 σ)
= min(0.21, 0.19) / (3 × 0.0268742)
= 2.3567
Process capability index C_{pk}
[23] cp = (USL - LSL) / (6 σ)
= (0.8 - 0.4) / (6 × 0.0268742)
= 2.4807
Process capability C_p
[24] dpmo = 10^6 × P(outside)
= 10^6 × Φ-tails at σ = 0.0268742
= 7.7755e-7
Predicted defects per million
[25] yield = 1 - dpmo / 10^6
= 1 - 7.7755e-7 / 10^6
= 100 %
Predicted yield
[26] u_{cpk} = cpk_{target} / cpk
= 1.33 / 2.3567
= 0.56436
Capability utilisation
[27] pareto = largest contributor (statistical)
= Housing bore depth 38.5 %; Bearing width 55.4 %; Spacer 6.15 %
= "Bearing width"
Contribution Pareto
[30] mc_{mean} = mean of 10000 samples (seed 1)
= Monte Carlo
= 0.60987 mm
Monte Carlo mean
[31] mc_{sigma} = sample standard deviation
= Monte Carlo
= 0.027228 mm
Monte Carlo σ
[32] mc_{p05} = 5 % quantile
= Monte Carlo
= 0.56589 mm
Monte Carlo 5 %
[33] mc_{p95} = 95 % quantile
= Monte Carlo
= 0.65462 mm
Monte Carlo 95 %
[34] mc_{yield} = samples inside the requirement
= 10000 of 10000
= 100 %
Monte Carlo yield

5. Results

SYMBOL	VALUE	DESCRIPTION
x_{nom}	0.5000 mm	Nominal result
x_{mean}	0.6100 mm	Mean result
t_{wc}	0.1300 mm	Worst-case tolerance (±)
t_{rss}	0.08062 mm	RSS tolerance (±)
σ	0.02687 mm	Predicted σ
$x_{wc,min}$	0.4800 mm	Worst-case minimum
$x_{wc,max}$	0.7400 mm	Worst-case maximum
t_{mrss}	0.1209 mm	Modified RSS tolerance (±)
m_{min}	0.08000 mm	Worst-case margin to the minimum
m_{max}	0.06000 mm	Worst-case margin to the maximum
cp	2.4807	C_p
cpk	2.3567	C_{pk}
dpmo	7.7755e-7	DPMO (predicted)
yield	100 %	Yield (predicted)
mc_{mean}	0.6099 mm	Monte Carlo mean
mc_{sigma}	0.02723 mm	Monte Carlo σ
mc_{p05}	0.5659 mm	Monte Carlo 5 % quantile
mc_{p95}	0.6546 mm	Monte Carlo 95 % quantile
mc_{yield}	100 %	Monte Carlo yield
u_{cpk}	0.56	Capability utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	C_{pk} against the requirement	1.33	2.3567	0.56	
PASS	Worst-case range against the requirement	0.48 … 0.74 mm	0.4 … 0.8 mm	—	

7. Applicability

WARNING contributors: there are fewer than four contributors. RSS and modified RSS are optimistic for so few. [few-contributors]

8. Verification

```
Template mech.tolerance-stack@2.3.0 Content hash 1b96b6f2b2031243 Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution. Technical Product Documentation and Specification. BS 8888. BSI.
2. American Society of Mechanical Engineers. Dimensioning and Tolerancing. ASME Y14.5. ASME.

3. International Organization for Standardization. General Tolerances—Part 1: Tolerances for Linear and Angular Dimensions Without Individual Tolerance Indications. ISO 2768-1. ISO.